

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway Associates, EliteUS Software Solutions Aug 2024 - Present | Edison, NJ

- Built large-scale recommender/ranking systems using Python, SQL, ML models to personalize search results and improve relevance, processing 5M+ rows of user and interaction data, increasing 30% user engagement.
- Developed multimodal retrieval and understanding pipelines leveraging NLP and vision models to improve content matching and relevance across text and image-based queries.
- Applied LLMs (GPT-4, LLaMA, Gemini) and retrieval-augmented generation to enhance relevance scoring, content understanding, and response quality, improving accuracy by 30%+ while reducing latency by 25%.
- Designed and deployed scalable ML infrastructure using FastAPI/Flask, cloud endpoints, and CI workflows, enabling reliable training, evaluation, and real-time inference in production systems.

Data Science Intern, Harvest Software Solutions June 2023 - Sept 2023 | Jacksonville, Florida

- Engineered scalable ML pipelines and monitoring systems for large advertising datasets, increasing campaign visibility from 15% to 45% while ensuring reliability and performance.
- Built ML services and NLP APIs using XGBoost, H2O AutoML, and FastAPI, improving sentiment classification accuracy by 65% and campaign spend efficiency by 50%.
- Designed SQL/Python data processing and evaluation pipelines using hash maps, sliding windows, priority queues, batching strategies to aggregate user and content features, improving engagement quality by 40%.

Application Development Associate, Accenture June 2021 - Aug 2022 | Hyderabad, India

- Developed and optimized Python-based ETL and data processing pipelines using hash maps, sets, and indexed lookups to efficiently clean, deduplicate, transform large datasets, reducing 50% production issues.
- Implemented SQL workflows and joins with partitioning, indexing, and window functions, enabling efficient aggregation and analysis over millions of records in AWS Athena and Redshift.
- Applied batching, caching, and algorithmic optimizations for data validation and pipeline execution, improving system performance and reliability by 25%.

TECHNICAL SKILLS

Programming: Python, R, C++, SQL. **Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, FastAPI.
ML / AI : Ranking/relevance modeling, recommender systems, model deployment, evaluation, feature engineering.
GenAI & LLMs: LLM fine-tuning, RAG, Prompt Eng., semantic embeddings, multimodal modeling, vector databases.
Cloud & MLOps: PostgreSQL, Data Warehousing, AWS (Bedrock, Athena, Redshift), GCP, Azure, Pinecone, Chroma.
Deep Learning: Neural Networks, CNNs, LSTMs, YOLO, Faster R-CNN, GANs, Time-Series Forecasting.

EDUCATION

Rutgers University, Master of Science in Data Science Sept 2022 - May 2024 | New Brunswick, NJ

Personal Website: <https://jayanthdasamantharao.github.io/>

Andhra University B.Tech in Electrical and Electronics Engineering 2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Self-Driving Cars (Computer Vision): Built autonomous driving cars with vision pipeline using CNNs, YOLO, Faster R-CNN, achieving 95%+ object detection accuracy.

Multimodal Retrieval System (Production): Built LLM-assisted retrieval and ranking pipelines with embeddings, caching, and text + image understanding to improve relevance and response quality in production systems.

Bayesian Classification: Implemented Bayesian logistic regression in PyMC3 to predict diabetes status, incorporating informative priors, achieving 0.85 F1-score.

For additional projects and code samples: <https://github.com/JayanthDasamantharao>